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PIGY, a new plant *envelope*-class LTR retrotransposon

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Abstract Plant LTR retrotransposons of the *envelope* class define a new branch in the *Metaviridae* family. They differ from other LTR retrotransposons mainly by the presence of an additional ORF downstream of the *gag-pol* region which has been hypothesized to be equivalent to the *envelope* gene of retroviruses. Here we present a newly identified element from pea (*Pisum sativum*), named *PIGY*, that has all the features characteristic of this group of LTR retrotransposons. In addition to the potential coding sequence downstream of the *gag-pol* region, *PIGY* has a primer binding site complementary to $tRNA^{asp}$ and a polypurine tract with a TGGGG motif and is of large size (13,645 bp). The relationship between *PIGY* and other retrotransposons of the *env*-class was confirmed by a phylogenetic analysis of their reverse transcriptase domains. One distinctive feature of *PIGY* is that its *env*-like region is actually composed of two similar ORFs, each of which encodes a protein with similarity to the *Athila* envelope-like protein. *PIGY* is present in the pea genome in $1–5\times10^3$ copies and is transcriptionally active, suggesting that some of these elements may still be capable of active transposition. Another new *env*-class retrotransposon similar to *PIGY* was also identified among genomic sequences of *Medicago truncatula*.

Keywords LTR retrotransposon · Envelope · *Pisum sativum* · *Medicago truncatula*

Introduction

Retroelements comprise a group of diverse genetic entities that use reverse transcription in their replication cycle. One taxonomic branch of retroelements, *Retrovirales*, is characterized by the possession of genes that encode GAG (or an equivalent protein), reverse transcriptase (RT) and ribonuclease H (RNaseH) proteins and by direct terminal repeats which have an important role in replication (Boeke and Stoye 1997; Hull 2001; Wilhelm and Wilhelm 2001). It includes three groups containing five families: LTR retrotransposons (*Pseudoviridae* and *Metaviridae* families; their members are often referred to as *Ty1*/*copia* and *Ty3*/*gypsy*-like retrotransposons, respectively), pararetroviruses (*Hepadnaviridae* and *Caulimoviridae*) and retroviruses (*Retroviridae*) (Boeke and Stoye 1997; Hull 2001). Although all these families can be considered viruses (Boeke and Stoye 1997), only the last three are actually infectious. Most LTR retrotransposons are not infectious. They represent mobile elements present in all eukaryotic genomes studied so far (Hirochika and Hirochika 1993; Suoniemi et al. 1998; Kumekawa et al. 1999; Schmidt 1999; Arkhipova 2001; Bhattacharya et al. 2002; Deininger and Batzer 2002; Wöstemayer and Kreibich 2002). LTR retrotransposons are structurally very similar to retroviruses; they are flanked by two long terminal repeats (LTRs), contain a primer binding site (PBS) downstream of the left LTR and a polypurine tract (PPT) upstream of the right LTR, and possess *gag-pol* genes encoding proteins which ensure reverse transcription and integration into the genome. While the sequences of reverse transcriptases, RNases H, and integrases (IN) differ between LTR retrotransposons and retroviruses (Xiong and Eickbush 1990; Malik and Eickbush 1999, 2001), retroviruses were set apart mainly due to the presence of an envelope gene (*env*) which is critical for the dispersal of viral particles from cell to cell, thus allowing them to be infectious. However, recent findings show that retrotransposons can acquire an additional, *env*-like ORF, which enables them to become infectious as well (Malik et al. 2000).

Infectious members of the LTR retrotransposons have been described in insects, and are called *Errantiviruses* (Kim et al. 1994; Song et al. 1994). Additional elements

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that acquired an *env* gene from a viral source have been described in *Caenorhabditis elegans* (*Cer*) and *Ascaris lumbricoides* (*Ted*), although their infectious nature has yet to be experimentally proven (Malik et al. 2000). It has recently been shown that some plant LTR retrotransposons also contain an additional ORF downstream of the *gag-pol* coding region which might represent a potential *env* gene. To date, elements that may contain an *env* gene include the *Ty3/gypsy*-like elements *Cyclops-2* from *Pisum sativum* (Chavanne et al. 1998), *Calypso* from *Glycine max* (Wright and Voytas 2001), *Athila* from *Arabidopsis thalina* (Wright and Voytas 2001), *Bagy-2* from *Hordeum vulgare* (Vicient et al. 2001) and *Rigy-2* from *Oryza sativa* (Vicient et al. 2001), and the *Ty1/copia*-like elements *SIRE-1* from *G. max* (Laten et al. 1998) , *Endovir* from *A. thaliana* (Peterson-Burch et al. 2000) and *ToRTL1* from *Lycopersicon esculentum* (Laten 1999). Since all vertebrate retroviruses have a *gypsy*-like arrangement of the *pol* domains (Vogt 1997), the presence of the potential *env*-gene in both *Ty3/gypsy*-like and *Ty1/copia*-like retrotransposons is striking, and suggests that these elements evolved at least twice (Laten 1999). Phylogenetic analysis based on reverse transcriptases has shown that the *Ty3/gypsy*-like *env*-bearing retrotransposons form a distinct branch among other *gypsy*-like but *env*-lacking elements (Peterson-Burch et al. 2000; Vicient et al. 2001) and similar results have been obtained for the *copia*-like *env*-bearing elements (Laten 1999; Peterson-Burch and Voytas 2002). Unlike the *pol* gene, which displays a high degree of sequence conservation among different *Ty3/gypsy*-like or *Ty1/copia*-like elements, the putative *env* sequences show very little mutual similarity, but they all encode a putative transmembrane domain. This domain is also the most conserved feature of the otherwise divergent retroviral env proteins (Swanstrom and Wills 1997; Vogt et al. 1997; Lerat and Capy 1999). An additional notable similarity to retroviruses was found in the element *Bagy-2*, which uses the same mechanism to express its *env* gene by splicing its RNA such that most of the *gag-pol* region is removed (Vicient et al. 2001). Although RNA splicing has not been experimentally confirmed in other elements, potential splice sites have been predicted in some of them (Wright and Voytas 2001). On the other hand, *SIRE-1* uses stop-codon suppression to express its env-like protein (Havecker and Voytas 2003), indicating that splicing is not the only mechanism employed by plant *env*-bearing retrotransposons. Despite certain resemblances between plant retrotransposon and retroviral env-like proteins, the function of plant *env*-like ORFs is still controversial as it has not been experimentally proven. Therefore, the *env*-bearing elements are referred to as "*env*-class retrotransposons" or "endogenous retroviruses" (Vicient et al. 2001; Wright and Voytas 2001).

Despite the fact that elements of the *env*-class seem to be widespread among plants (Vicient et al. 2001), the number of elements described to date is very small, and all of these appear to have accumulated mutations which complicate their analysis. In addition, the absence of intact elements prevents experiments that could clarify the role of the *env*-like ORF in plant retrotransposons. It is therefore important that more elements with *env*-like genes are identified, as this should increase the chance of finding some that are intact or allow the construction of a functional element based on direct and reconstructed sequence data. In this paper, we describe *PIGY*, a new member of the *env*-class of retrotransposons in *P. sativum*, which contains two *env*-like ORFs, is abundant in the pea genome and is still transcriptionally active. Moreover, a close relative, *MEGY*, has been identified among sequences released by the *Medicago truncatula* Genome Sequencing Project.

Materials and methods

Plant material

Seeds from pea (*P. sativum* L. cv. Carrera) were obtained from the Plant Breeding Station at Boršov (Czech Republic). Seeds from other species of the Viciaceae were obtained from the legume seed bank maintained by Agritec (Šumperk, Czech Republic), or from the IPK (Gatersleben, Germany). Total genomic DNA was extracted from leaves as described by Dellaporta et al. (1983). DNA concentrations were determined using PicoGreen dye (Molecular Probes) according to the manufacturer's recommendations.

Sequence analysis

Basic sequence analysis was done using the Staden Package (Staden 1996) and program tools implemented at the Biology WorkBench server (http://workbench.sdsc.edu/). Dot-plot sequence comparisons were carried out using the Dotter program (Sonnhammer and Durbin 1995). Multiple sequence comparisons were performed using CLUSTALW (Thompson et al. 1994). Homology searches were performed using BLAST and FASTA (Pearson and Lipman 1988; Altschul et al. 1997; http://www.ebi.ac.uk/Tools/homology.html). The search for conserved protein domains was done using RPS-Blast (Marchler-Bauer et al. 2003; http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi). Sequences of tRNAs used for identification of PBSs were obtained from the *A. thaliana* tRNA database (Lowe and Eddy 1997; http://rna.wustl.edu/tRNAdb/). Splice site analysis was performed on the NetGene2 server (Hebsgaard et al. 1996; http://www.cbs.dtu.dk/services/NetGene2/). Potential transmembrane domains were predicted using Tmpred (Hofman and Stoffel 1993; http://www.ch.embnet.org/software/TMPRED_form.html).

Copy number estimation

To estimate the copy number of *PIGY*, dilutions of pea genomic DNA (10^3–10^5 C) and *PIGY-1* DNA (10^4–10^9 copies) were quantitatively dot-blotted on Hybond N+ membranes (Amersham). The amounts of DNA blotted

were calculated assuming that 1 pg of dsDNA is equivalent to 9.65×10^8 bp (Bennett and Smith 1976) and that the size of the haploid genome of pea is 4.43 pg (Baranyi and Greilhuber 1996). Element copy number was estimated based on a comparison of signal intensities after sequential hybridizations with probes prepared from four different regions of *PIGY-1* (Fig. 1a). In addition to dot-blot hybridizations, copy number was also estimated based on the numbers of hybridizing clones identified in phage and cosmid libraries, assuming that average insert sizes were 15 and 30 kb, respectively, and that each positive clone contained only a single copy of *PIGY*. Detection of *PIGY* sequences in genomes of other plant species was done using dot-blots bearing amounts of DNA equivalent to 10^5 copies of the haploid genome of the species of interest and 10^8 copies of *PIGY* DNA. Probe labeling and detection were performed using the AlkPhos direct kit (Amersham). All hybridizations were carried out at 65°C according to the manufacturer's recommendations.

RNA isolation and RT-PCR

Tissues used for RNA isolation (leaf, root, flower) were harvested from plants cultivated in pots under a 15-h light/9-h dark cycle at 22 and 18°C, respectively. Alternatively, leaves were wounded by abrading the surface with emery paper, and the leaves were harvested 2 days later. Protoplasts were prepared from leaves that had been cut into small pieces and incubated in an enzyme mixture (1.375% Cellulase R10, 0.2% Macerozym R10 in 0.5 M sucrose, 5 mM $CaCl_2$, 10 mM potassium MES) for 15 h at room temperature. Total RNA was isolated using the Total RNA Isolation Kit (Ambion). All RNA samples were treated with RNase-free DNase (GIBCO BRL) to remove any contaminating DNA. Reverse transcription was carried out using the SuperScript II Reverse Transcription Kit (GIBCO BRL) by a random priming method according to manufacturer's recom-

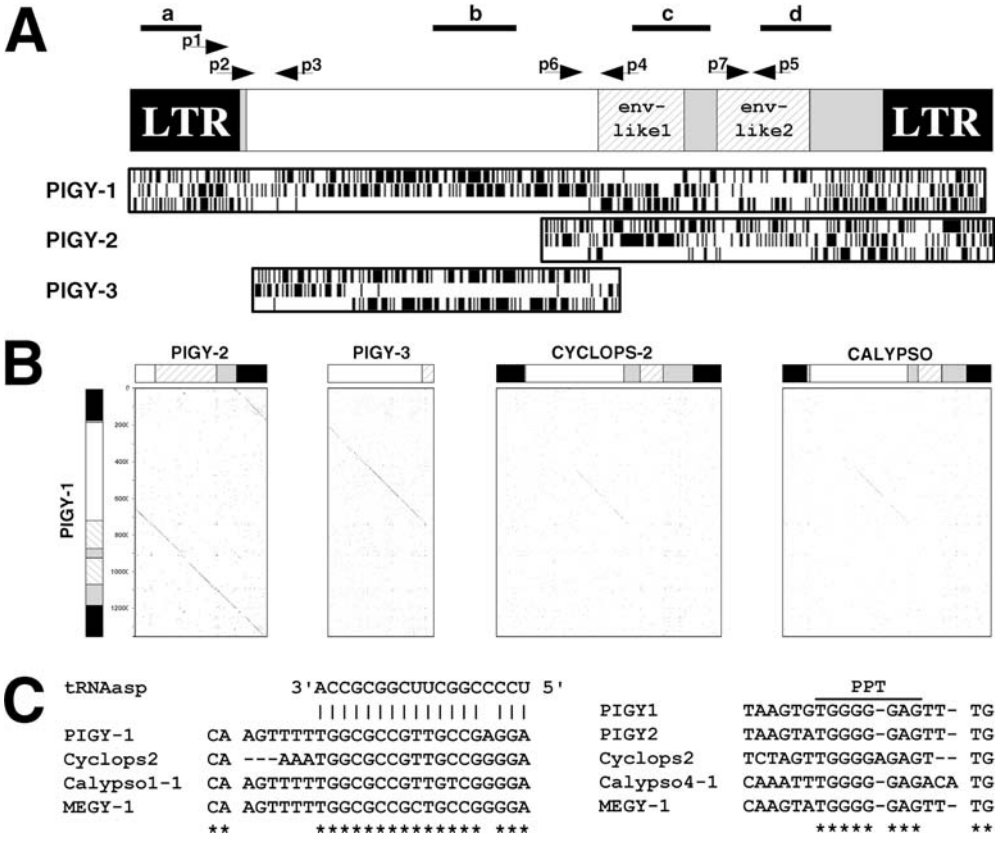


Fig. 1 Schematic representation of the retrotransposon *PIGY*. **a** Structure of the full-length *PIGY* element. The LTRs are indicated as *black boxes*, the putative non-coding regions (5′ and 3′ untranslated regions, the region between the *env*-like ORFs) by *gray boxes*, and the *gag-pol* gene by the *white box*, while the *cross-hatched boxes* represent the additional potentially coding regions designated *env*-like1 and *env*-like2. A description of individual domains that are typical of polyproteins encoded by *gag-pol* genes (*white box*) is given in Table 1. The positions of the ORFs *env*-like1 and *env*-like2, although they contain some stop codons, are defined as follows: *PIGY-1*: positions 7393–8757 and 9287–10840; *PIGY-2*: 915–2291 and 2766–4211; *PIGY-3*: 5200–5796. The *arrows* and *overlines* show the positions and orientation of the primers used for RT-PCR (p1–p7) and the positions of the probes used in hybridization experiments (a–d). The plots *below* the diagram show ORFs in all three frames as regions without *vertical lines* (which represent stop codons). **b** Dot-plots made by the Dotter program (Sonnhammer and Durbin 1995) displaying regions of similarity between *PIGY-1* and *PIGY-2*, *PIGY-3*, *Cyclops-2* or *Calypso* (a consensus sequence created from *Calypso1-1* and *Calypso 3-1*). Individual structural regions of the elements are marked in the same way as in **a**. **c** Homology of the PBS and PPT among *PIGY-1*, *MEGY-1*, *Cyclops-2*, and *Calypso1-1* elements

mendations, using 0.5 μg of template RNA. The RT-PCR reaction mix (25 μl) consisted of 1×PCR buffer, 0.2 μM dNTPs, 0.2 mM primers, 1.5 mM $MgCl_2$, 1 U of Platinum *Taq* polymerase (Invitrogen) and 2.5 ng of reverse transcribed RNA or an equal amount of RNA as a negative control. The following primers were used in the RT-PCR experiments: p1 (5′ CAACAGAGCG-AGAGTTTGAG 3′), p2 (5′ TGGCGCCGTTGCCG-GGGA 3′), p3 (5′ TCCTCTAAACCATGGTGTGG 3′), p4 (5′ CCATRCGATCAAWATGTCCC 3′), p5 (5′ AGAGGCGGCGTTTACCTT 3′), p6 (5′ ACYAGTG-GACAAGTGGAAGT 3′) and p7 (5′ YACCAGGTA-CCCNGATTTA 3′). The reaction was carried out for 35 cycles of 30 s at 94°C, 50 s at 55°C and 1–3 min at 72°C, preceded by an initial denaturation step (3 min at 94°C) and followed by a final extension step (10 min at 72°C). Reaction products were resolved by agarose gel electrophoresis.

Fluorescence in situ hybridization (FISH)

FISH was performed on isolated chromosomes prepared as described by Gualberti et al. (1996) and centrifuged onto slides using a Hettich centrifuge equipped with cytospin chambers. The probe for *PIGY* was labeled with biotin-16-dUTP (Boehringer Mannheim) by random priming. To distinguish between individual chromosomes, we simultaneously used a probe for PisTR-B tandem repeats labeled with Alexa Fluor-568-5-dUTP (Neumann et al. 2001). Treatment of slides before hybridization and the compositions of the hybridization mix and washing solutions were as described by Neumann et al. (2001). Hybridization was carried out at 25°C for 16 h and followed by washing at 32° for 10 min. Detection of the biotinylated probe was performed as described by Leitch et al. (1994) using fluorescein–avidin and biotinylated anti-avidin (Vector). Chromosomes were counterstained with DAPI (4′,6′-diamidino-2-phenylindole) and examined using a Nikon Eclipse 600 microscope equipped with appropriate filter sets. The images were captured with a CCD camera and analyzed using Lucia software (Laboratory Imaging).

Results

Sequence analysis of the genomic clone Ps-cos16 from pea (GenBank Accession No. AY299398) revealed that, in addition to the previously described retroelement *Ogre* (Neumann et al. 2003), it contained another element defined by two LTRs flanking an internal sequence that could potentially code for retroelement proteins (Fig. 1a). This coding region displayed sequence similarity to several plant retrotransposons, including the previously described pea LTR retrotransposon *Cyclops-2* (Chavanne et al. 1998). As this similarity was limited to the *pol* region (Fig. 1b), which is relatively conserved among different retrotransposons, we decided to consider this element a new retrotransposon and named it *PIGY-1*. Partial sequences similar to *PIGY-1*, designated *PIGY-2* and *PIGY-3*, were also identified in two other pea genomic clones, Ps-phage13 (AY299394) and Ps-cos14 (AY299397), respectively.

Features of the *PIGY* retrotransposon

PIGY is almost completely represented by the *PIGY-1* sequence (13,519 bp) which lacks only 126 bp at the 3′ end of its right LTR. The other two *PIGY* sequences are truncated to a greater extent, either due to their position at the border of the cloned fragment (*PIGY-2*, 7096 bp) or to insertions of sequences of an unknown origin (*PIGY-3*, 5796 bp) (Fig. 1a). *PIGY-2* and *PIGY-3* show 77.1 and 65.7% similarity to *PIGY-1*, respectively.

The LTRs of *PIGY-1* differ by only 28 base substitutions in 1618 bp, and are therefore 98.3% identical to each other. A PBS complementary to the 3′ end of $tRNA^{asp}$ (Lowe and Eddy 1997) was identified 7 bp downstream of the left LTR. It differs from $tRNA^{asp}$ and from the PBS of the related elements *Cyclops-2* and *Calypso* by one base substitution (Fig. 1c). A putative PPT was found 3 bp upstream of the right LTR and contains a TGGGG motif which is also fully conserved in *Cyclops-2* and *Calypso* (Fig. 1c). A putative *gag-pol* coding region within *PIGY-1* starts 110 bp downstream of the 3′ end of the left LTR and is 5400 bp in length. The whole region could be translated, in one frame, into a polyprotein composed of 1800 amino acids, although there are three stop codons within the *gag* region. A search for characteristic motifs within the protein sequence identified all the typical domains, i.e. GAG, PR, RT, RNaseH and IN (Table 1). As the RT coding domain precedes the IN coding domain, *PIGY* can be regarded as an LTR retrotransposon of the *Ty3*/*gypsy* type. Also like *Cyclops-2, PIGY* has an additional potential coding region downstream of the *gag-pol* ORF, which in the case of *Cyclops-2* and other *env*-class retrotransposons has been hypothesized to be an *env* ORF. However, this region of *PIGY* does not show any significant similarity in DNA sequence to the related *env*-class retrotransposons *Cyclops-2* and *Calypso* (Fig. 1b).

Similarity to other *env*-class retrotransposons

To determine if *PIGY* actually belongs to the group of *env*-class retrotransposons we carried out a phylogenetic analysis with ClustalW, employing the alignment of various RT sequences constructed by Vicient et al. (2001) (obtained from the EMBL Alignment Database: Accession No. ALIGN_000602) with *PIGY* sequences added. The analysis clustered *PIGY* and all *env*-class retrotransposons on the same branch (bootstrap value = 100%) showing that *PIGY* is indeed a new member of this group (Fig. 2). This classification was further confirmed by the similarity of the RT domain of *PIGY*

Table 1 Domains found within protein sequences of *PIGY-1*

Protein[a]	Domain	Score	position	Notes
Polyprotein	GAG (pfam03732)	$1e^{-15}$	138–230	Zinc finger motif CX2CX4HX4C (322–335 aa)
	PR	–	650–655	LCDLRA motif
	RT (pfam00078)	$9e^{-24}$	927–1114	
	RNaseH	–	1212–1321	DEAD motif
	IN (pfam00665)	$1e^{-22}$	1496–1648	
Env-like1	pfam03078	$2e^{-11}$	7546–8754	Env-like protein of retrotransposon *Athila*
Env-like2	pfam03078	$6e^{-16}$	9587–10525	Env-like protein of retrotransposon *Athila*

[a]Protein sequences deduced from potentially coding regions found in the *PIGY-1* element were used to search the Conserved Domain Database (CDD) using RPS-Blast (Marchler-Bauer et al. 2003). The sequence of the polyprotein (1800 residues) was obtained by translating the *gag-pol* gene (1854–7253 bp). Although the CDD contains accessions for all characteristic retrotransposon domains, only three of them provided good scores. The remaining two domains, PR and RNaseH, had to be identified using protein motifs described by Wright and Voytas (2001) and Malik and Eickbush (2001), respectively. Env-like1 and 2 represent protein sequences encoded within the additional coding region present downstream of the *gag-pol* gene. Unlike the case for the polyprotein, the positions of the pfam03078 domain found within these proteins are given with respect to their nucleotide sequences

to RT coding sequences of the *env*-class of retrotransposons previously amplified by PCR from a number of species (Vicient et al. 2001; Wright and Voytas 2001).

A comparison of *PIGY* sequences with other retrotransposons of the *gypsy*-like *env*-class, including *Cyclops-2*, *Calypso*, *Bagy-2*, and *Rigy-2*, revealed that regions conserved between these elements were limited to the *pol* region, the PBS complementary to the 3′ end of $tRNA^{asp}$, and the PPT (Fig. 1b, c; Vicient et al. 2001; Wright and Voytas 2001). Other parts, even those encoding for gag and potential env proteins, differed considerably from one another. A search for other *env*-class retrotransposons using the *PIGY-1* nucleotide sequence as the query found a similar element in *M. truncatula*, which we named *MEGY-1* (AC146683.9: positions 116515–125617). This element is 9103 bp long and contains an ORF coding for a polyprotein, but not the *env*-like ORF. In contrast to other *PIGY*-related *env*-class elements, *MEGY-1* shares significant similarity (66%) with *PIGY-1* over the whole *gag-pol* coding region (Fig. 3). A PBS homologous to $tRNA^{asp}$ and a PPT highly similar to that of *PIGY* were found as well (Fig. 1c). As we had identified many additional sequences similar to *MEGY-1*in the *M. truncatula* genome, we attempted to identify among them *MEGY-1*-related elements that had *env*-like ORFs, using TBLASTN with the *PIGY-1* env-like1 protein sequence as a query. Although a total of nine such sequences were found, only three of them (*MEGY-2*, Accession No. AC140551.7: positions 23003–30894; *MEGY-3*, Acc. No. AC145023.4: 15604–27912, including an internal insertion at 18816–24045; *MEGY-4*, Acc. No. AC139747.15: 61168–104039, including two insertions at 66507–96138 and 97729–103750) were flanked by two LTRs which defined the borders of the elements. All three elements had large deletions within the *pol* ORF, indicating that they are not capable of autonomous transposition.

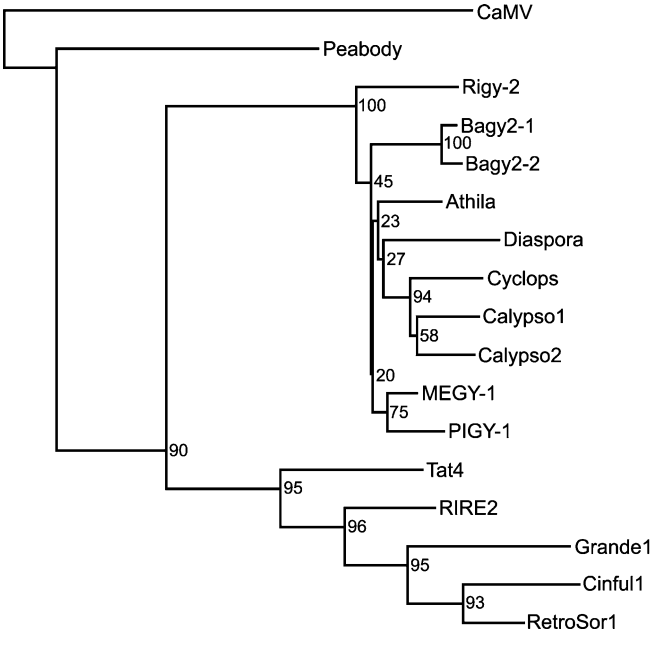


Fig. 2 Phylogenetic tree based on the analysis of reverse transcriptase domains. RT-sequences derived from *PIGY-1* and *MEGY-1* were included in the alignment ALIGN_000602 of Vicient et al. (2001) and the phylogenetic analysis was performed using ClustalW. Only a part of the tree is shown, because other branches were almost identical to the original tree and can be seen in the aforementioned paper. The numbers on branches are bootstrap values (%) from 1000 replicates

Env-like ORFs

One remarkable feature of *PIGY* is the presence of two additional ORFs, *env*-like1 and *env*-like2 (1365–1377 and 1446–1554 bp in length, respectively) downstream of the *gag-pol* ORF (Fig. 1a). The first *env*-like ORF is either separated from *gag-pol* by a short sequence (137 and 199 bp in *PIGY-1* and *PIGY-2*, respectively) or overlaps by 28 bp with the *pol* ORF (*PIGY-3*). The *env*-like ORFs, which occur either in the same frame (*PIGY-2*) or in two different frames (*PIGY-1*), are separated from each other by a region containing stop codons in all frames that is 529 and 474 bp in length in *PIGY-1* and *PIGY-2*, respectively (Fig. 1a). While the *env*-like ORFs in *PIGY-1* and *PIGY-2* showed high similarity to one another, the *PIGY-3 env*-like1 ORF

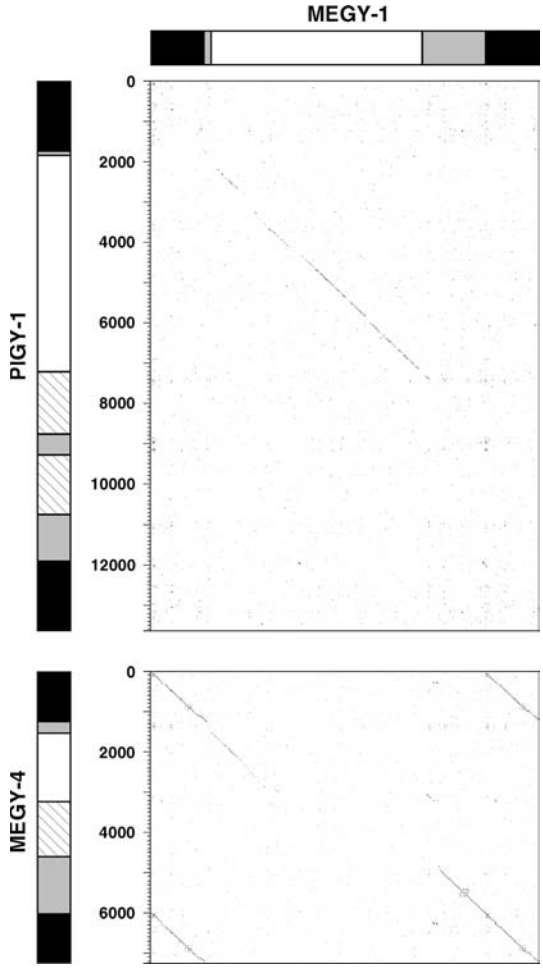


Fig. 3 Dot-plot comparison of *PIGY* and *MEGY* DNA sequences. Individual structural regions of the elements are marked in the same way as in Fig. 1a

was markedly different in DNA sequence (Fig. 1b and data not shown). When these ORFs were translated into protein sequences we were able to identify a region of similarity within *env*-like1 and *env*-like2 sequences (Fig. 4) that was also found in the *PIGY-3* sequence. As only the positions themselves are insufficient to justify the designation of *PIGY env*-like ORFs as *env* genes, we tested whether their predicted products shared other features characteristic of env proteins. A search for transmembrane domains performed using the TMpred program (Hofmann and Stoffel 1993) found two domains in the *env*-like1 and one domain in the *env*-like2 product (Fig. 4). Using RPS-BLAST, we also tested whether proteins encoded by either of these ORFs showed similarity to any retroviral *env* entries in the Conserved Domain Database. Significant similarity (scores of $2e^{-6}$–$2e^{-13}$) was found in the case of both translated *env*-like sequences to pfam03078 (Table 1), which represents an env-like protein encoded by the retrotransposon *Athila* from *A. thaliana*. Moreover, it is noteworthy that although the transmembrane domains predicted for the potential *PIGY* and *Athila* env-like proteins had different protein sequences, their positions were the same in both cases (Fig. 4). On the other hand, a comparison of the potential *PIGY* env proteins with corresponding proteins from other plant *env*-class retrotransposons did not reveal any significant similarities.

Retroviral *env*-genes are translated from spliced RNAs from which most of the *gag-pol* coding region has been removed. If this were the case for *PIGY*, there should be acceptor splice sites within, or in the vicinity of, its *env*-like ORFs. The prediction of splice sites, performed at the NetGene2 server, provided several splice sites which differed in position in all three *PIGY* sequences (Table 2). However, most of these splice sites seemed unlikely to be active, because their use would result in the splicing of excessively long regions out of these ORFs.

As in the case of *PIGY*, env-like protein sequences derived from *MEGY* showed similarity to the *Athila* env-like protein (pfam03078; score $1e^{-5}$–$8e^{-6}$). However, they did not have any predicted transmembrane domain (Fig. 4). A search for splice sites could not be performed because no full-length *MEGY* sequence was available. We hypothesized that deletions within the *pol* gene of the elements *MEGY-2, 3* and *4* could have originated from splicing, and therefore tested whether the 5′ ends of these deletions corresponded to any of the predicted donor sites in *MEGY-1*. Although there were several such sites, none of them could be associated with the 5′ end of the deleted region (data not shown). However, it is not clear whether this is because sequence divergence prevented a precise alignment of donor sites with the 5' end of the deleted region or indicates that the deletions arose by a mechanism other than splicing.

Copy number, genomic organization and distribution of *PIGY* elements in other species

To estimate the copy number of *PIGY* elements in the pea genome we used two approaches, based on dot-blot and colony/plaque-blot hybridizations (see Materials and methods). To avoid copy number overestimation caused by cross-hybridization of *PIGY*-DNA with related retrotransposons (e.g. *Cyclops*), probes from four different regions of *PIGY*, including the LTR, *pol*, *env*-like1 and *env*-like2 segments (probes a, b, c and d, respectively; Fig. 1a) were used under stringent conditions. Both approaches gave similar results irrespective of the probe used, showing that the copy number of *PIGY* in the pea genome is between 1×10^3 and 5×10^3. As revealed by fluorescence in situ hybridization with probes a and b carried out on isolated metaphase chromosomes, individual copies of *PIGY* were dispersed along all seven chromosomes of the pea karyotype (Fig. 5).

All four probes used for copy number estimation were also hybridized to DNA from several other species of the Viciaceae. The results revealed that *PIGY* is

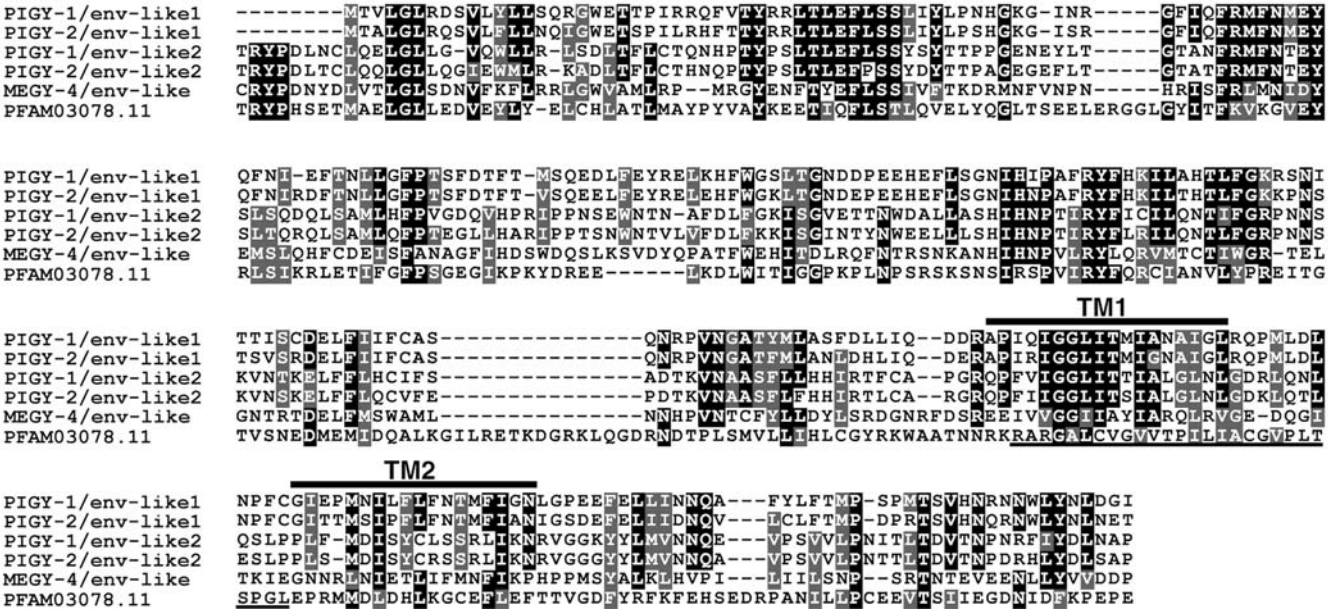


Fig. 4 Multiple alignment of env-like protein sequences. All *PIGY* env-like protein sequences could be aligned within the region that also showed similarity to *Athila* and *MEGY* env-like protein sequences. The *PIGY-3* protein sequence is not shown because it includes only a short part of the region. Transmembrane domains (*TM*) predicted for *PIGY env*-like proteins are *overlined*. TM1 was predicted for both env-like proteins, whereas TM2 was predicted only for *env*-like1. Note that position of the TM1 domain is nearly the same as in *Athila* (*underlined* region; Wright and Voytas 2001). The *MEGY* sequence does not have any predicted TM domain

present in *Pisum* and *Vicia* species but absent from other genera. However, there were differences depending on the probes used. While probes for LTR and *pol* regions hybridized with DNA from several species, both probes for *env*-like ORFs were specific for *Pisum* (Table 3).

Transcriptional activity

To test whether *PIGY* elements were transcribed in pea, we performed RT-PCR with the primers p2 and p3, which were designed to amplify the region delimited by the PBS and the 5′ part of the *gag*-coding region (Fig. 1a). RNA used for the experiment was isolated from leaves, roots, flowers, wounded leaves and protoplasts. Reaction products of the expected size were detected in all five RNA samples, as well as in a sample containing genomic DNA as the template, which was included as a positive control (Fig. 6a). Negative controls (omitting the reverse-transcription step) yielded no product, thus excluding the possibility of DNA contamination of the RNA samples. RT-PCR products were cloned, and four to six clones from each tissue were sequenced to confirm that the amplified sequences were derived from transcripts of *PIGY*. The degree of similarity of these sequences to *PIGY* varied between 90 and 96%, indicating that they were derived from different elements belonging to the *PIGY* family. This result was confirmed by RT-PCR with the primer pairs p1 and p3, p4 and p6, and p5 and p7, which

Table 2 Acceptor splice site prediction

Name[a]	Position	Confidence	5′ Intron^exon 3′
PIGY-1	7059	0.92	TTATTCCCAG^GAAAGCTACG
	8076	0.90	**TGCTTCCCAG^AACCGTCCAG**
	8278	0.71	ATGTTTATAG^GAAACCTTGG
	9757	0.86	CACCTTGGAG^TTTTTAAGTT
PIGY-2	975	0.65	TTTTCTCTAG^TTAACACTAA
	989*	0.97	CACTAACCAG^ATGGGACATT
	1151	0.78	GCTGAATCAG^ATAGGATGGG
	1586	0.76	**TGCTTCCCAG^AACCGTCCAG**
	1643	0.69	CCTTATCCAG^GATGAGCGAG
PIGY-3	4614	0.67	GTGGAGTTAG^GCATAGAGTA
	5432*	0.95	GTGTTTTCAG^GATCCTTTTC
	5456*	0.96	ACCTTTGCAG^GATGCAGAAC

[a]With one exception (shown in *bold*), all predicted acceptor splice sites differ among the three *PIGY* sequences. In addition, most of the splice sites seem unlikely to be active because their positions imply splicing of excessively long regions within these ORFs (see positions of *env*-like ORFs in Fig. 1). The *asterisks* indicate splice sites that give high confidence scores and are in suitable positions

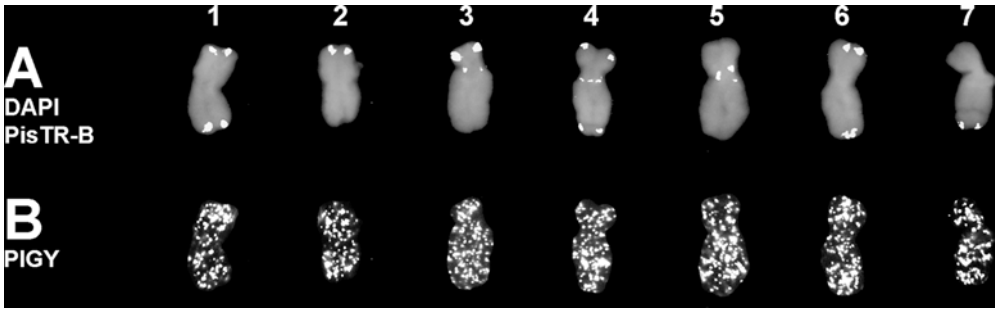


Fig. 5 Localization of *PIGY* sequences on mitotic chromosomes using FISH. **a** Seven chromosomes of the pea karyotype stained with DAPI (*gray*). Individual chromosome types were discriminated by hybridization with the probe for PisTR-B (*white*) (Neumann et al. 2001). **b** Hybridization signals corresponding to the LTR of PIGY (probe a, Fig. 1a) obtained on the same set of chromosomes

were designed from less conserved regions of *PIGY* including the LTR and both env-like ORFs (Fig. 6b, c, and data not shown). The transcriptional activity of *PIGY* allowed us to test for splicing of its RNA, which is a prerequisite for expression of the *env* gene in retroviruses. The forward primer p2 was used in combination with either of the reverse primers p4 and p5. However, attempts to amplify spliced RNA failed, although all of these primers were successfully used in the previous experiments (see Fig. 1 for primer positions).

BLASTN was used to search the EST database to determine if *MEGY* is transcribed in the *M. truncatula* genome. Although four EST sequences with significant scores ($<e^{-137}$) were identified, only one of them (BE322689) was similar to MEGY along its whole length, whereas the remaining three sequences (CB065414, BI265348 and BM815455) showed only partial similarity and might represent read-through transcripts or transcripts derived from partially different elements. All of these ESTs originated from treated plants. Thus these findings do not allow us to conclude that *MEGY* elements are transcriptionally active under normal physiological conditions.

Discussion

The recent discovery of *env*-class retrotransposons in plants is of great importance because it raises the possibility that infectious retrotransposons exist in plants. Although one member of this retrotransposon group has already been described in *P. sativum* (*Cyclops*; Chavanne et al. 1998), its similarity to *PIGY* is limited to the *pol* gene, which is conserved in sequence among all *env*-class retrotransposons (Fig. 1b; Wright and Voytas 2001). Therefore, *PIGY* can be regarded as a new member of the *env*-class of retrotransposons. Compared to *Cyclops-2*, *PIGY-1* seems to be a more recent insertion because its LTRs share greater similarity (98.3 vs. 95.9%; this work; Chavanne et al. 1998). In addition, we presume that the *PIGY-2* insertion is relatively recent as well, because it was found to be inserted within the *Ogre* retrotransposon, which is thought to have been amplified in the pea genome quite recently (Neumann et al. 2003).

PIGY has all the structural features characteristic of the plant *Ty3/gypsy*-like *env*-class retrotransposon family. It is large in size (13,645 bp) and has a PBS complementary to $tRNA^{asp}$, a PPT containing a TGGGG motif and, most importantly, the additional coding region downstream of the *gag-pol* gene (Wright and Voytas 2001). However, its DNA sequence similarity to other members of this group of retrotransposons is limited to the part of the coding region comprising the *pol* gene; the remaining domains (even those encoding *gag* and *env*-like ORFs) are markedly

Table 3 Distribution of *PIGY* sequences among selected *Viciaceae* species

Probe	Species[a]													
	PE	VF	VG	VH	VM	VN	VP	VS	CA	GM	LA	PV	Vun	MT
LTR (a)	++	+	+	++	++	++	++	+	–	–	–	–	–	–
pol (b)	++	++	++	++	++	++	++	++	–	–	–	–	–	–
env-like1 (c)	++	–	–	–	–	–	–	–	–	–	–	–	–	–
env-like2 (d)	++	–	–	–	–	–	–	–	–	–	–	–	–	–

[a]Genomic DNA from *Pisum elatius* (PE), *Vicia faba* (VF), *V. grandiflora* (VG), *V. hybrida* (VH), *V. melanops* (VM), *V. narbonensis* (VN), *V. pannonica* (VP), *V. sativa* (VS), *Cicer arietinum* (CA), *G. max* (GM), *Lupinus angustifolium* (LA), *Phaseolus vulgaris* (PV), *Vigna unguiculata* (Vun) and *M. truncatula* (MT). was carried out with probes prepared from different parts of *PIGY*, listed in the first column (a, b, c and d; Fig. 1A). Signal intensities were divided into three categories: >1000 copies/C (++), <1000 copies/C (+) and not detected (–). The results show that *PIGY* sequences are present in only some species from the family Viciaceae, and only some parts of *PIGY* are conserved in these species

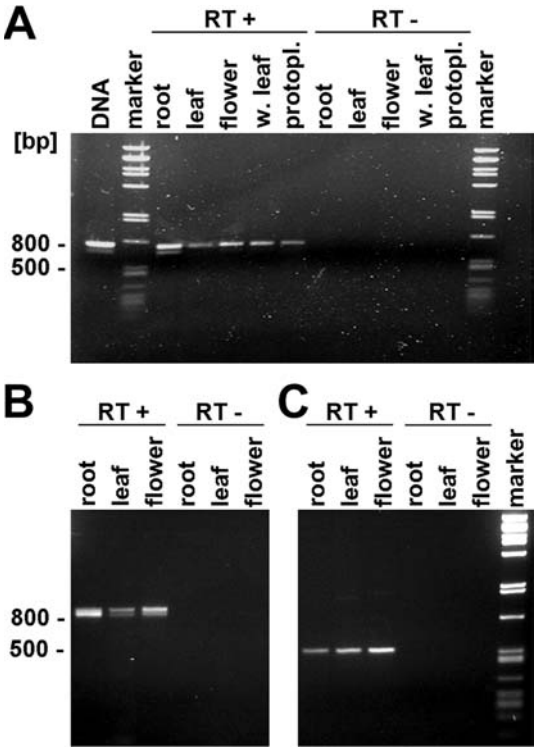


Fig. 6 RT-PCR. The reactions were carried out with total RNA isolated from tissues harvested either from plants grown under normal conditions (root, leaf, flower), or from tissues or cells that had been subjected to stress conditions (wounding of leaves and protoplasting). Two sets of otherwise identical reactions differing only by the addition (*RT+*) or omission (*RT−*) of the reverse transcription step were carried out in order to check for possible false-positive results caused by contamination with genomic DNA. *Panels* a, b, and c show the results of reactions performed with primers p2 and p3, p4 and p6, and p5 and p7, respectively

different. This sequence heterogeneity was observed among all *env*-class retrotransposons reported to date (our observations; Wright and Voytas 2001). Despite this, phylogenetic analysis based on sequences of RT domains shows that these elements belong to one clade which is well separated from clades containing *env*-lacking elements (this work; Vicient et al. 2001).

A distinctive feature of *PIGY* is the presence of two *env*-like ORFs, designated *env*-like1 and 2. The protein sequences deduced from these ORFs show significant localized similarity between *env*-like1 and 2 (Fig. 4). This segment also shows significant similarity to the env-like protein (pfam03078) of the retrotransposon *Athila* (Wright and Voytas 2001). On the other hand, env-like protein sequences derived from other *env*-class retrotransposons seem to be very divergent and could not be aligned either with *PIGY* or *Athila* env-like proteins. The relatively high sequence divergence is evident even among different members of the *Athila* set (Wright and Voytas 2001) as well as among the three *PIGY* elements themselves. The rapid evolution of *env*-like ORFs is confirmed by the negative results of our hybridization experiments with genomic DNA from related species belonging to the Viciaceae (Table 3). As env-like protein sequences derived from *Cyclops-2* and *Calypso* could be aligned with one another (29% identity; Peterson-Burch et al. 2000) their divergence from *PIGY* is striking. This may suggest that subgroups of *env*-class retrotransposons acquired their *env*-like ORFs from different sources.

The only common feature shared by env-like proteins of all *env*-class retrotransposons, including those of *PIGY*, is the presence of a transmembrane domain, which is thought to be one of the most important characteristics within otherwise divergent env-like proteins (Swanstrom and Wills 1997; Vogt 1997; Lerat and Capy 1999). On the other hand, Krogh et al. (2001) reported that transmembrane domains are present in a broad spectrum (20–30%) of proteins, indicating that this domain itself cannot serve as an unambiguous criterion for env proteins. In addition, env-like proteins derived from *MEGY* elements have no predicted transmembrane domain, although their sequences are similar to those deduced from *PIGY* sequences, which makes the importance of this domain questionable. It should be kept in mind that none of the *MEGY* elements bearing an *env*-like ORF described here was intact, and that the absence of the domain could alternatively be explained as a result of accumulated mutations. Support for the hypothesis that the *env*-like ORFs of plant *env*-class retrotransposons could be *env* genes was provided by Vicient et al. (2001), who demonstrated splicing of *Bagy-2* transcripts to produce subgenomic mRNAs in a manner similar to that seen in retroviruses (Rabson and Graves 1997). Splicing of *PIGY* and of other *env*-class retrotransposons has not been proven experimentally, although the transcripts contain predicted splice sites. It might also be speculated that the large deletions in the *pol* region of *MEGY-2, 3* and *4* are a consequence of splicing; however, sequence differences between these elements and *MEGY-1* did not allow us to confirm this hypothesis unambiguously (data not shown). Some env-like proteins contain consensus protease cleavage sites, N-glycosylation sites and leucine zipper domains that are also typical for retroviral envelope proteins (Vicient et al. 2001). In spite of this, a role for env-like proteins is not yet clear. As *MEGY-1* lacks an *env*-like ORF, it is obvious that this ORF is not necessary for transposition of *env*-class elements, although it is likely that this ORF has other important functions.

The transcriptional activity of *PIGY* is not surprising, in light of the fact that RNA of *env*-class retrotransposons has also been detected in several other species (Vicient et al. 2001). However, it does suggest that *PIGY* may still be capable of transposition within the pea genome. This provides an encouraging perspective for future research aimed at elucidating the role and the regulation of expression of *env*-like ORFs.

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